SIMATS ENGINEERING

CSA0705 - COMPUTER NETWORKS

STUDENT NAME : BALAJI A

REGISTER NO : 1911260245

DEPARTMENT : B.TECH COMPUTER SCIENCE ENGINEERING

ACADEMIC YEAR : 2026

FACULTY NAME : KUMARAN G

NATURE OF ASSIGNMENT : INDIVIDUAL

1. Ethernet Port Report Analysis

The shop-floor PC has a detected 100 Mb/s half-duplex link. Auto-negotiation is disabled. The interface also reports receive errors, late collisions and CRC errors.

Field	Reported value	Meaning
Speed	100 Mb/s	The link speed is 100 megabits per second.
Duplex	Half	The device can transmit or receive, but not both at the same time.
Auto-negotiation	Off	The link settings are not being automatically negotiated.
Link detected	Yes	A physical network link is detected.
rx_errors	2381	The interface has recorded received-packet errors.
late_collisions	412	Collisions were detected later than the normal collision window.
crc_errors	1907	Received Ethernet frames failed the CRC/FCS error check.

Late collision

A late collision occurs when an Ethernet collision is detected later than the normal collision-detection window. It is abnormal on a modern full-duplex switched link and may indicate a duplex mismatch or a physical-layer problem.

CRC error

CRC means Cyclic Redundancy Check. Ethernet uses it to detect corrupted frames. A CRC error means a received frame failed its frame-check sequence. Possible causes include a damaged cable, poor connectors, interference, a faulty network interface card (NIC) or switch port, or a link-configuration problem.

What the combination suggests

The high late-collision and CRC-error counts suggest a link problem. A duplex mismatch or inconsistent port configuration is a likely cause to investigate. The PC report alone does not prove the switch's exact setting, so both ends of the link must be checked.

2. Corrective Configuration – Windows

For a modern gigabit Ethernet connection, the recommended approach is to enable auto-negotiation on both the PC and the switch port so that both ends agree on the link settings. Auto-negotiation is required for 1000BASE-T gigabit Ethernet.

- Open Device Manager in Windows.
- Expand Network adapters.
- Right-click the Ethernet adapter (for example, the Realtek Gaming GbE Family Controller) and select Properties.
- Open the Advanced tab.
- Find Speed & Duplex. The exact label may vary by network-adapter driver.
- Select Auto Negotiation (or the equivalent automatic setting) and click OK.
- Ensure that the connected switch port is also configured for auto-negotiation.

A forced setting such as 100 Mb/s Full Duplex should only be used if the switch port is also manually set to exactly the same speed and duplex. Do not force only one end.

Verification

After changing the setting, reopen the adapter properties and verify that Auto Negotiation is selected. A real negotiated speed/duplex result and error-counter check require an active Ethernet connection. Because the submitted laptop evidence is from Wi-Fi, no Ethernet correction is claimed as completed on this laptop.

3. MAC Address Analysis

01:00:5e:0a:01:02

This is a multicast (group) MAC address, not an individual unicast device address. The 01:00:5e prefix is used for IPv4 multicast Ethernet addresses. A multicast frame can be delivered to a group of interested devices.

3c:52:82:4e:11:07

The first three bytes, 3C:52:82, form the OUI (Organizationally Unique Identifier). The OUI identifies the organization or manufacturer to which that MAC-address block was assigned.

4. Jumbo Frames for the CCTV Storage Server

Jumbo frames may help when transferring large CCTV video files. A common Ethernet MTU is 1500 bytes, while a jumbo-frame MTU is commonly around 9000 bytes. Larger frames reduce the number of frames and per-frame processing overhead, potentially improving transfer efficiency. The actual benefit depends on the equipment and workload.

For jumbo frames to work reliably, a compatible MTU must be supported and configured end-to-end on:

- The CCTV camera, recorder or other sending device, where applicable.
- The storage server's network interface card (NIC).
- Every switch and intermediate network device along the path.

If one device on the path cannot handle the configured frame size, frames may be dropped or communication may fail.

5. Alternative Concept Analysis – Ring vs Star

Point	Ring topology	Star topology
Connection	Industrial switches connect in a loop.	Devices or access switches connect to a central switch.
One cable cut	A fast ring-recovery protocol can redirect traffic through the other side of the ring.	The device or branch connected by the cut cable loses its connection.
Recovery time	Typically tens of milliseconds with a configured fast-recovery protocol; exact time depends on the protocol and configuration.	No automatic recovery for a single-connected device; service returns after the cable is repaired.
Affected devices	Devices generally remain reachable after ring reconvergence, although a brief interruption may occur.	Other branches normally continue working; the cut branch is affected.

The question does not name a ring-redundancy protocol, so an exact recovery time cannot be stated. Fast recovery depends on the protocol and its configuration. A star network also depends on the central switch remaining operational.

6. Practical Evidence – Windows Screenshots

Evidence note: The screenshots below document the Windows Wi-Fi connection used on the student's laptop. They are supporting evidence of the active network connection only; they do not show Ethernet duplex settings, Ethernet error counters, or prove that an Ethernet correction was applied. The laptop has no Ethernet cable connected during these captures.

Screenshot 1 – Active Wi-Fi configuration (ipconfig /all)

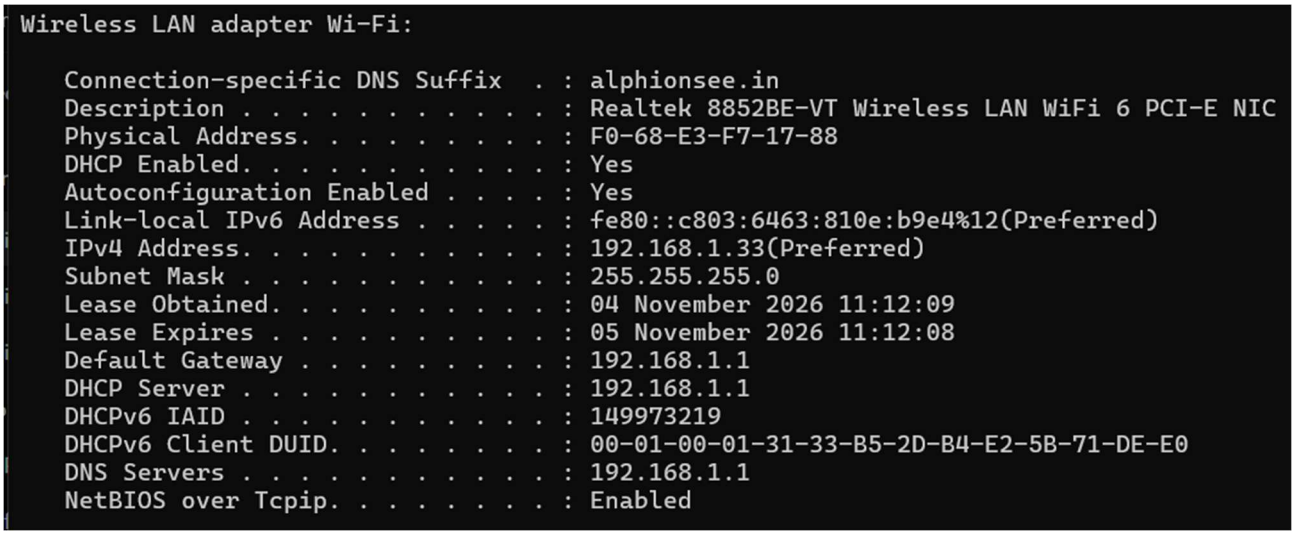


Figure 1: Active Wireless LAN adapter details shown by the Windows ipconfig /all command.

Screenshot 2 – Wi-Fi connection status

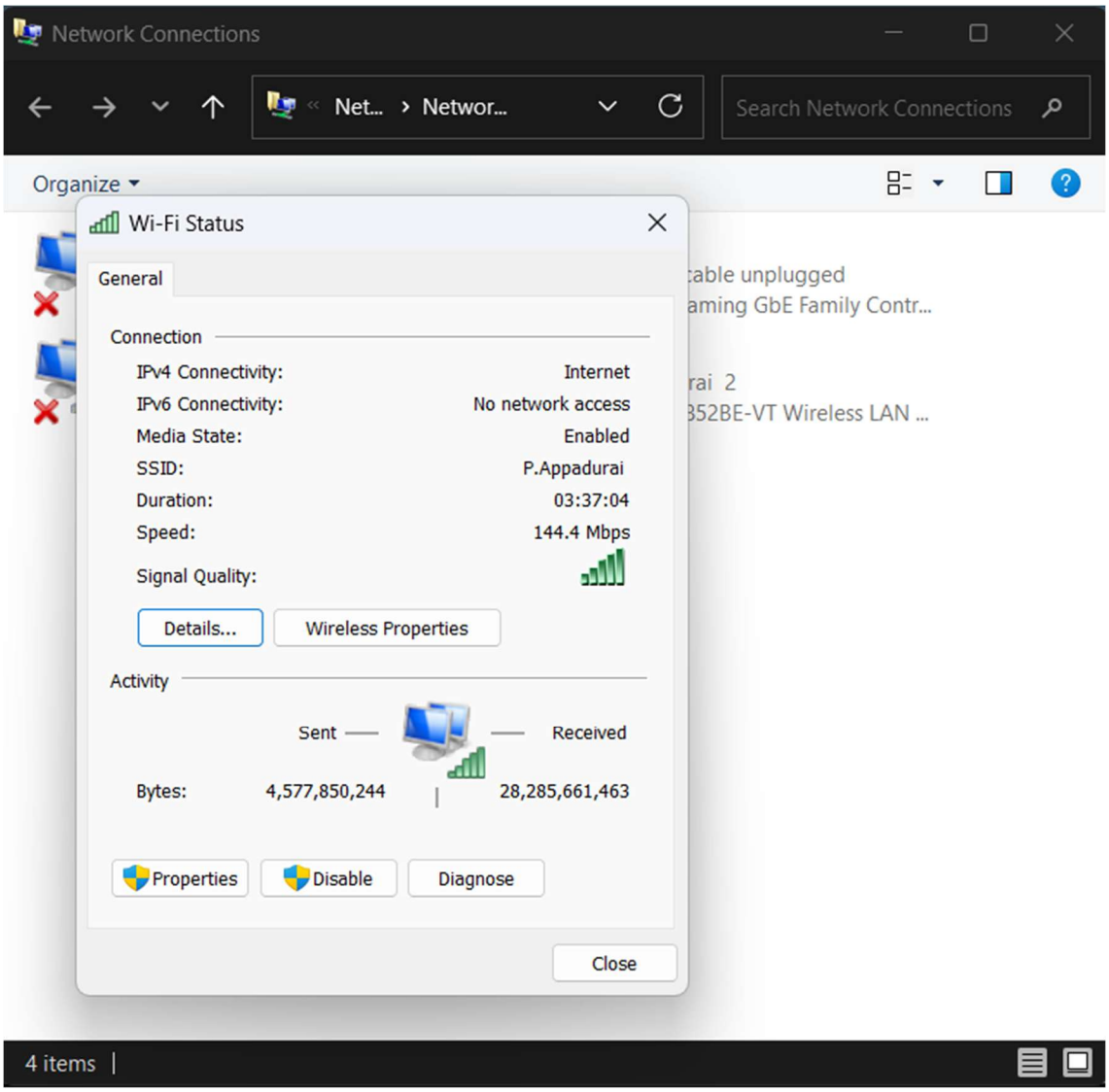


Figure 2: Windows Wi-Fi Status window showing the connected SSID, link speed and signal quality.